VATSALYA HOSPITAL

Central Pharmacy & Clinical Inventory ERP

OFFICIAL STANDARD OPERATING MANUAL (SOP)

Vatsalya Pharmacy ERP Software Operating Handbook

Complete Step-by-Step Operator Manual for Pharmacists & Storekeepers

A comprehensive operational manual covering retail counter sales, FEFO automated batch dispensing, E-Prescription queue synchronization, IPD ward medication indents, Medication Administration Record (eMAR) reconciliation, inventory master catalog, near-expiry alerts, quarantine & statutory disposal, P2P procurement (PO → GRN → Invoices → Payments), Schedule H/H1 registers, and GST tax filings.

SYSTEM VERSION	DOCUMENT ID	PHARMACY FACILITY
Pharmacy ERP v2.0 Enterprise	VH-PHARM-SOP-2026-01	Vatsalya Central Pharmacy
TARGET AUDIENCE	**REGULATORY STANDARD**	**EFFECTIVE DATE**
Pharmacists, Cashiers, Storekeepers, Auditors	Drugs & Cosmetics Act 1940 / GST	October 2026

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> ⓘ **REGULATORY COMPLIANCE MANDATE**
>
> Vatsalya Pharmacy operates under strict compliance with the Drugs and Cosmetics Act (1940), Rules (1945), and Good Pharmacy Practice (GPP). All dispensed prescription medicines, batch records, cold chain temperatures, and Schedule H/H1 registers must be audited daily.

01 System Architecture & POS Hardware Setup

1.1 System Architecture & Dual Database Sync

Vatsalya Pharmacy ERP is engineered as a dedicated high-throughput retail and clinical pharmacy suite. It uses a dual-connection architecture:

- **Primary Pharmacy Engine:** Manages real-time POS barcode scanning, rapid batch stock lookups, supplier procurement, and inventory ledgers.
- **MediPro Hospital DB Interoperability:** Synchronizes live with `hospital_db` to read patient registrations (UHID), doctor lists, admitted IPD patient censuses, and pending e-prescriptions.

COMPONENT	SPECIFICATION	RECOMMENDED CONFIGURATION
Access URL (Local)	`http://localhost/Vatsalya%20Hospital/Pharmacy/login.php`	Dedicated desktop shortcut on POS counters
Database Integration	Shared `hospital_db` MySQL Schema	UTF8MB4 encoding, Asia/Kolkata timezone
POS Barcode Scanner	USB 1D/2D High-Speed Laser / Image Scanner	HID Keyboard Mode with automatic [ENTER] suffix
Receipt Printer	80mm (3-inch) Direct Thermal POS Printer	Baud Rate 9600 / USB, auto paper-cutter enabled
Laser / Inkjet Printer	A4 / A5 Flatbed Laser Printer	Used for GST Invoices and Supplier Purchase Orders
Cash Drawer	RJ11 interface connected to Thermal Printer	Auto-kick pulse trigger upon cash invoice print

1.2 Cold-Chain Storage Hardware Protocols

Biologics, insulins, vaccines, and sera must be stored in calibrated medical refrigerators between **+2°C and +8°C**. Digital data loggers must be inspected twice daily at 09:00 AM and 05:00 PM, and recorded in the Pharmacy Cold Chain Register.

02 User Roles, Shift Sessions & Cash Float

2.1 Pharmacy User Roles & Permissions Matrix

USER ROLE	PERMISSIONS & DAILY RESPONSIBILITIES	RESTRICTED CAPABILITIES
ADMIN	Full system control, product master management, pricing changes, supplier payments, tax configurations, and audit trail inspection.	None (Full Access)
PHARMACIST	Dispensing prescriptions, OPD counter sales, IPD ward indents, sales returns, stock adjustments, and Schedule H/H1 registers.	Cannot modify tax slabs or delete audit logs.
CASHIER	Counter billing, payment collection (Cash, UPI, Card), shift cash drawer settlement, and thermal receipt printing.	Cannot create new medicines or adjust purchase prices.
STOREKEEPER	Purchase orders, GRN inward goods receiving, rack placement, expiry tracking, and quarantine movements.	Cannot override sales prices or issue refunds.

2.2 Daily Shift Opening & Cash Reconciliation SOP

1. **Open Daily Shift Session**
   Cashier logs in at shift start (e.g. 08:00 AM) and enters the opening cash drawer float (e.g., ₹2,000 in coins and small denominations).

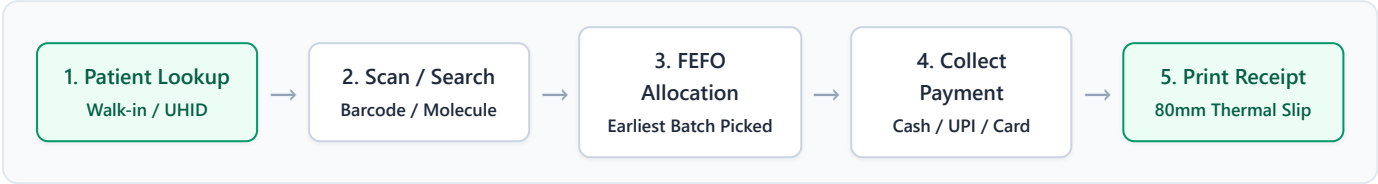
2. **Continuous Real-Time Counter Billing**
   System records every transaction categorized by tender type: Cash, Dynamic UPI QR, Card POS, or Credit.

3. **End-of-Shift Reconciliation**
   Cashier clicks **Close Shift**. System calculates expected cash vs physically counted cash, flags any variance (±), and prints the Shift Settlement Slip for cash drop into the hospital safe.

03 Point of Sale (POS) Counter Operations & FEFO

3.1 High-Speed Retail POS Sales Workflow (modules/sales/counter.php)

The OPD Counter is optimized for ultra-fast, barcode-driven checkouts during peak hospital hours:



Step-by-Step Counter Sale Execution:

1. **Identify Customer / Patient:**
   - For Hospital Patients: Type UHID (e.g. `VH00142`) or mobile number in the search bar. Patient details, age, gender, and doctor auto-populate.
   - For Walk-in Customers: Press `F2` or leave default as "*Walk-in Customer*".
2. **Scan or Search Medicine:**
   - Barcode Mode: Aim scanner at product barcode. The item, batch, and MRP instantly add to cart.
   - Manual Search: Type brand name (e.g. "*Augmentin 625*") or generic molecule (e.g. "*Amoxicillin Clavulanate*"). The dropdown shows available packs, current stock, and shelf location (e.g. `Rack B, Shelf 3`).
3. **Enter Quantity & Unit Conversion:** Enter quantity. For loose strip sales (e.g. 6 tablets from a 10-tablet strip), the system automatically divides the strip price proportionally.
4. **Select Referring Doctor:** Select hospital specialist from dropdown or type external prescribing physician's name (required for Schedule H drugs).
5. **Tender Payment & Checkout:**
   - `Cash:` Enter cash received (e.g. ₹500 for a ₹340 bill). System displays exact change to return (₹160) and triggers cash drawer pulse.
   - `UPI / QR Code:` Displays dynamic UPI payment QR code for instant scan.
   - `Card POS:` Enter debit/credit card approval authorization code.
6. Click **Print Invoice (F9)**: Generates GST-compliant thermal receipt with batch numbers, expiry dates, and hospital tax breakdown.

3.2 Automated FEFO (First-Expiry-First-Out) Engine

> ✓ **AUTOMATED EXPIRY WASTE PREVENTION**
> The FEFO algorithm automatically selects the batch with the earliest valid expiry date from available stock. This prevents older batches from sitting at the back of shelves until they expire.

MEDICINE	BATCH NO	STOCK QTY	EXPIRY DATE	FEFO ENGINE PRIORITY
Paracetamol 650mg	**B-8901**	120 Tabs	Nov 2026	PRIORITY 1 (AUTO-SELECTED)
Paracetamol 650mg	**B-9144**	300 Tabs	Mar 2027	PRIORITY 2 (QUEUED)
Paracetamol 650mg	**B-9502**	500 Tabs	Jan 2028	PRIORITY 3 (QUEUED)

05 Doctor E-Prescriptions & IPD Ward Indents

5.1 E-Prescription Dispensing Queue (modules/sales/prescription.php)

When doctors finalize consultations in MediPro HMS, prescriptions appear in real-time in the Pharmacy Dispensing Queue:

1. **Load Pending Prescription**
   Scan patient token slip or enter UHID. The screen pulls the complete electronic prescription written by the consulting doctor.

2. **Verify Stock & Generic Alternatives**
   System compares prescribed drugs against live warehouse inventory. If the exact brand is out of stock, click **Suggest Generic Equivalent** to dispense an identical molecule with the same strength.

3. **One-Click Cart Conversion & Directions**
   Click **Convert to Bill**. All items, batches, and dosage instructions (e.g. "*1 Tab After Dinner for 5 Days*") transfer directly into the checkout cart.

5.2 Inpatient (IPD) Ward Medication Indents (modules/ipd/indents.php)

Inpatient wards (ICU, Male Ward, Female Ward, NICU) requisition medications through digital ward indents:



Key Operational Features of IPD Dispensing:

- **No Counter Cash Collection:** When dispensing against an active IPD patient, the transaction automatically posts directly as an itemized debit to the patient's centralized hospital bill (`hospital_db.bills`).
- **Picking List Printout:** Pharmacist prints a picking slip listing item name, required quantity, allocated batch number, and physical shelf/rack location for rapid assembly.
- **Ward Delivery Confirmation:** The delivering pharmacy technician and receiving ward nurse verify and sign the physical delivery challan.

07 Inpatient MAR Reconciliation & Returns

7.1 Medication Administration Record (MAR) Synchronization

To maintain complete audit traceability between pharmacy stock and patient administration, the MAR synchronization module (`modules/ipd/mar.php`) links dispensed drugs with nursing administration:

CLINICAL EVENT	SYSTEM SYNCHRONIZATION	REGULATORY TRACEABILITY
Dispensed by Pharmacy	Stock decremented from Pharmacy; debited to IPD patient bill.	Batch and expiry recorded in dispensing audit ledger.
Administered by Nurse	Nurse marks dose administered on ward tablet with timestamp.	Status updated to `ADMINISTERED`.
Omitted / Withheld Dose	Nurse logs clinical reason (e.g. *Patient vomited, surgery fasting*).	Flags in Administration Variance Report for pharmacist review.

7.2 Inpatient Ward Returns & Credit Notes (modules/ipd/reconciliation.php)

Upon patient discharge, recovery, or physician modification of treatment, unused medications must be returned from the ward to the Central Pharmacy:

1. **Receive Ward Return Consignment**
   Ward nurse brings unused drugs with return slip. Pharmacist selects patient's active IPD file in `reconciliation.php`.

2. **Physical Blister Inspection**
   Inspect blister seals. Open bottles, cut strips, or cold-chain breaches cannot be accepted.

3. **Automatic Bill Credit & Inventory Restoration**
   Click **Accept Return**. The returned quantities automatically restore to their respective batch stock ledgers, and a credit adjustment is posted to the patient's hospital bill.

7.3 Customer Retail Sales Returns (modules/sales/returns.php)

- **Invoice Verification:** Enter the original Invoice Number. The system displays all items sold, original sale date, and cashier ID.
- **Batch Authenticity Check:** Verify that the returned item's batch matches the batch printed on the original receipt.
- **Issue Credit / Refund:** Print Return Credit Voucher and return cash or credit against new purchases.

09 Medicine Master Catalog & Batch Management

9.1 Adding & Editing Products (modules/inventory/products.php)

Every SKU stocked in Vatsalya Central Pharmacy must be properly classified in the Product Master:

FORM FIELD	FIELD TYPE	OPERATIONAL GUIDELINES & STATUTORY STANDARDS
Brand / Trade Name	Text	Commercial brand name as marketed (e.g., *Pan-D, Augmentin 625, Monocef 1g*).
Generic Molecule	Text	Active chemical composition (e.g., *Pantoprazole 40mg + Domperidone 30mg*).
Dosage Form	Dropdown	Tablet, Capsule, Syrup, Suspension, Injection, IV Infusion, Cream, Eye Drops, Inhaler.
Manufacturer	Select / Text	Pharmaceutical company (e.g., *Sun Pharma, Cipla, Alkem, Torrent, Mankind*).
HSN Code	Numeric (8 digits)	Standard GST HSN Code (e.g., `30049099` for allopathic formulations).
GST Slab	Dropdown	`0%` (Exempt/Life Saving), `5%` (Critical), `12%` (Standard formulations), `18%`.
Storage Shelf Location	Text (Rack/Shelf)	Physical location (e.g., `Rack-A4 / Shelf-2 / Bin-08`) for instant counter picking.
Schedule Category	Select	`Normal OTC`, `Schedule H`, `Schedule H1`, `Schedule X` (Narcotics).

9.2 Batch Stock Management & Pricing Structure

> ⓘ **PHARMACEUTICAL PRICING HIERARCHY**
> **PTS (Price to Stockist) → PTR (Price to Retailer / Purchase Price) → MRP (Maximum Retail Price).**

Under the DPCO (Drugs Price Control Order), no medicine can ever be sold above its printed MRP. When entering inward stock:

1. **Batch Number:** Exact alphanumeric batch code printed on the foil (e.g. `AC-2041`).
2. **Manufacturing (MFG) & Expiry (EXP) Dates:** Enter month and year (e.g. `04/2026` to `03/2028`).
3. **MRP:** Stamped retail price per unit or per strip.
4. **Purchase Rate (PTR):** Net purchase cost from vendor before tax.
5. **Reorder Level:** Minimum inventory threshold (e.g., 50 strips). When stock falls below this level, the system triggers an automatic purchase alert.

11 Expiry Control, Quarantine & Stock Auditing

11.1 Automated Expiry Monitoring Zones (modules/inventory/expiry.php)

The system categorizes expiring stock into color-coded proactive alert zones:

ALERT ZONE	EXPIRY TIMELINE	SYSTEM COLOR CODE	REQUIRED PHARMACIST ACTION PROTOCOL
Yellow Notice	Within 90 Days	NOTICE ZONE	Flag for fast-track dispensing or consult supplier for credit exchange.
Orange Warning	Within 60 Days	WARNING ZONE	Separate physically to Near-Expiry Tray; contact vendor for debit note return.
Red Critical	Within 30 Days	CRITICAL ALERT	Final return deadline to authorized pharmaceutical distributor.
Expired	Past Expiry Date	EXPIRED / LOCKED	System automatically blocks POS scanning; move to Quarantine Lockup.

11.2 Quarantine Lockup & Safe Disposal Protocol

> ⚠ **ZERO TOLERANCE POLICY: EXPIRED MEDICINE DISPENSING**
> Dispensing expired medication is a severe medical hazard and criminal offense under the Drugs & Cosmetics Act. Any expired item is locked out of POS billing.

1. **Move to Quarantine (modules/inventory/quarantine.php):** Select the expired batch and click **Transfer to Quarantine**. The stock transfers from active shelves to the secure quarantine cage.
2. **Regulatory Disposal Register (modules/inventory/disposals.php):** Unreturnable expired medicines are destroyed following biomedical waste disposal standards:
   - Record disposal method (Incineration / Encapsulation / Chemical destruction).
   - Mandatory sign-offs by Chief Pharmacist and Medical Superintendent.
   - System writes off inventory value and issues a statutory Disposal Certificate.

11.3 Physical Stock Auditing & Adjustments (modules/inventory/adjustments.php)

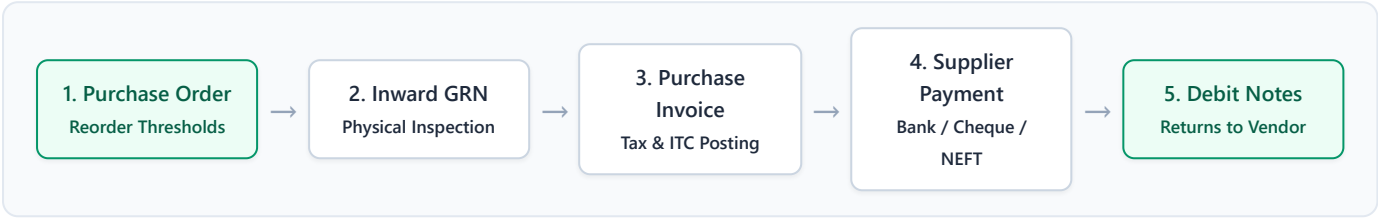
Monthly cycle counts ensure physical warehouse bins match digital records:

- **Stock Adjustment Reasons:** Mandatory reason codes: *Physical Audit Discrepancy*, *Breakage / Dropped Bottle*, *Liquid Spillage*, *Supplier Sampling*, or *Shrinkage*.
- **Supervisor Authorization:** Any stock adjustment above ₹500 requires Admin authentication.

13 Procurement Lifecycle (PO → GRN → Invoices → Payments)

13.1 Complete Procure-to-Pay (P2P) Cycle

Vatsalya Pharmacy utilizes a structured procurement lifecycle ensuring zero supply stock-outs:



Step-by-Step Procurement Procedures:

1. **Purchase Orders (modules/purchases/orders.php):** Review items below reorder level. System auto-calculates recommended order quantities based on 30-day sales velocity. Select approved vendor, approve PO, and email PDF order to distributor.
2. **Goods Received Note — GRN (modules/purchases/grn.php):** When shipment arrives at receiving dock:
   - Check delivery challan against physical boxes.
   - Verify batch numbers, manufacturing dates, and ensure expiry is at least 18 months away.
   - Record Free / Bonus Goods (e.g. *10 Boxes + 1 Free*). The system automatically averages unit purchase costs.
3. **Purchase Invoices (modules/purchases/invoices.php):** Enter supplier's tax invoice number, bill date, and transport LR number. System posts Input Tax Credit (ITC) CGST/SGST and updates warehouse stock immediately.
4. **Supplier Payments & Ledger (modules/purchases/payments.php):** Record cheque, RTGS/NEFT, or bank transfer payments. Allocate payment against outstanding bills and maintain supplier aging balances.
5. **Purchase Returns / Debit Notes (modules/purchases/returns.php):** For damaged, short-expiry, or recalled consignments, generate a Debit Note to deduct value from the vendor's payable balance.

15 Statutory Compliance, GST Reports & Administration

15.1 Statutory Drug Registers (Schedule H & H1 Compliance)

> ⚠ **STATUTORY INSPECTION MANDATE: DRUGS & COSMETICS RULES**
> Schedule H and H1 medications (third-generation cephalosporins, carbapenems, fluoroquinolones, psychotropics) must be recorded in the official inspection register for a minimum of 3 years.

Every sale of a scheduled drug automatically logs into the Statutory Register (`modules/reports/sales.php`):

- Date of Dispensing & Patient Full Name, Address, and Contact Mobile.
- Name and Medical Council Registration Number of the Prescribing Doctor.
- Drug Name, Manufacturer, Dispensed Batch Number, and Exact Quantity Sold.
- Signature / Electronic Authentication of the Registered Pharmacist.

15.2 Pharmacy Financial & Tax Reports (modules/reports/)

REPORT NAME	KEY METRICS CAPTURED	AUDITING PURPOSE
Daily Sales Summary	Gross sales, discounts given, net revenue, Cash vs UPI vs Card breakdown.	Daily cashier cash-drop reconciliation and financial closing.
Inventory Valuation	Stock value at Purchase Cost (PTR) vs Stock value at MRP.	Monthly balance sheet asset valuation and margin auditing.
Batch Traceability	Supplier Invoice → GRN → Warehouse Bin → Patient Sales Invoices.	360° rapid product recall within 15 minutes of manufacturer notice.
GSTR-1 Sales Report	B2B vs B2C taxable turnover, HSN summary, CGST / SGST liability.	Monthly GST return filing with Government GST portal.
GSTR-2 Purchase ITC	Vendor GSTIN, invoice value, eligible Input Tax Credit claims.	Reconciliation with supplier GSTR-2B tax statements.

15.3 Daily Operational Checklist for Pharmacists

- ☑ **08:00 AM:** Open POS shift, verify cash float, check refrigerator temperature (2°C–8°C).
- ☑ **12:00 PM:** Review Near-Expiry dashboard (30-day critical zone); clear pending IPD indents.
- ☑ **04:00 PM:** Check items below Reorder Level; compile purchase order requirements.
- ☑ **08:00 PM:** Perform evening cash drop reconciliation; print daily GST sales summary; lock schedule drug register.